



Examiners' Report Lead Examiner Feedback

January 2021

Pearson BTEC Nationals
In Applied Science (31617H)
Unit 1: Principles and Applications of Science I

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January 2021

Publications Code 31617H_2101_ER

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Introduction

Biology

Structure and functions of cells and tissues.

Medical professionals need to understand the structure and workings of cells. They build on this knowledge to understand how the body stays healthy as well as the symptoms and causes of some diseases. This allows them to diagnose and treat illnesses. The study of bacterial prokaryotic cells gives an understanding of how some other diseases are caused and can be treated.

Introduction to the Overall Performance of the Unit

Structure and functions of cells and tissues.

There was evidence that time management during the exam is improving with more learners attempting all questions. Exam technique and the understanding of the command verbs is continuing to improve. It is beneficial for learners to appreciate the requirement of each of the command verbs so that they can target their response and provide appropriate credit worthy answers. Explain is the command verb which seems least understood. Learners' explanations require a justification/exemplification of a point. The answer must contain some element of reasoning/justification – this can include mathematical explanations. Centres need to fully prepare learners for the exam by practising exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response in order to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary. Learners should show that they understand the relationship between the structure and function of cells and tissues. Learners still struggle to recall definitions and find it difficult to provide links between different related aspects of the specification.

Introduction

Chemistry

This was the seventh sitting of the level 3 Applied Science Unit 1 Chemistry section and the fourth time the paper has been sat separately to the other disciplines that makes up the unit. It is pleasing to see that learners are still accessing the paper well with few blank spaces seen and learners achieving the full range of marks across the whole of the paper.

Introduction to the Overall Performance of the Unit

Learners that were successful this year were able use scientific vocabulary and terminology and were able to apply their knowledge of Chemistry to different situations and contexts. They were able to perform calculations in an unknown scenario, those that did well structured their working well to come to the correct answer. Learners that performed well also were able to draw diagrams that were labelled accurately using the correct technical language.

To improve further, centres should ensure that learners are aware of the difference between key terms such as property and use, chemical and physical properties. They should also be aware of the difference in command words such as give, state, describe and explain and how these types of questions should be approached.

Introduction

Physics

The examination was set to allow learners to show their knowledge and understanding of waves in the context of communication and to use their knowledge to interpret and analyse the use of waves in different applications.

Introduction to the Overall Performance of the Unit

The majority of learners have shown improvement in mathematical skills, most now show substitution into a simple equation and give an evaluation. More complex equations requiring rearrangement and functions such as square roots are still found to be challenging but most learners followed the procedure of showing substitution, attempting a rearrangement and evaluating and could therefore gain marks. Although learners are very familiar with the quantities used to describe transverse waves and may recognise a longitudinal wave when shown on a 'slinky' they are less familiar with the longitudinal wave shown as a series of compressions and rarefactions and were less likely to be unable to identify a wavelength for a longitudinal wave. Learners were generally able to recognise total internal reflection shown in an optical fibre but need to improve their understanding of the purpose and effect of adding cladding to the fibres. Also, the way apparatus was used to produce a stationary wave on a wire was not well understood although there was frequent reference to the wave being reflected at the fixed end.

Learners need to appreciate the need to know the order of the waves in the electromagnetic spectrum in terms of increasing frequency and to realise the relationship between frequency and wavelength, so that the same order applies to decreasing wavelength. There is also some confusion between the properties of the different electromagnetic waves used for communication and the use of terms such as 'multiplexing' need to be clearly understood. The diffraction grating and its use in producing an emission spectrum using the light from a sodium lamp was much better understood and a significant number of learners were able to explain this in terms of constructive and destructive interference. However, some learners misunderstood the

question and described how light of different wavelengths was produced by the lamp rather than how the diffraction grating produced the emission spectrum.

Individual Questions

Biology

Q1(a)(i)

This was a 1 mark multiple choice question, which required the learners to recall the parts of the ECG trace.

This response was awarded 1 mark.

(a) (i) Identify the P-wave in Figure 1.



Q1(a)(ii)

This was a 1 mark question. Very few identified QRS, but many were able to mention sinus wave to obtain the mark. A commonly observed incorrect response was hyperpolarisation.

This response was awarded 0 marks. This is a very common incorrect response.

(ii) Name the ECG complex that  represents in Figure 1.

(1)

~~depolarisation~~ repolarisation

This response was awarded 1 mark.

The mark can just be awarded as the learner has clearly made an attempt at spelling depolarisation. Depolarisation is an acceptable response and we do not have to include ventricular to award the mark. Their attempt at spelling does not give another related word and so can be accepted. We can also accept contraction and systole without ventricular.

(ii) Name the ECG complex that  represents in Figure 1.

(1)

Depolarisation.

Q01(b)(i)

This was a 1 mark multiple choice question assessing the learners knowledge about the names of tissues.

This response was awarded 1 mark.

(i) Identify the type of tissue that forms the lining of the inside of arteries.

- ☐ A connective
- ☒ B endothelial
- ☐ C muscular
- ☐ D nervous

Q1(b)(ii)

This was a 3 mark question. Many learners provided the correct answers, but in the incorrect order. The most common marking point awarded was for C, however many learners lost a mark by answering 'clotted'. There was some confusion between 'atheroma' and 'atherosclerosis'.

This response was awarded 0 marks.

Incorrect missing words have been given for all 3. Unfortunately, the learner has given a correct response for A in space for B - this is a common mistake! For C the learner has stated thick - and so is not awarded a mark. The statement for C states "reduced blood flow because the inside of the artery becomes....C...." and so the statement is referring to the lumen not the artery wall. The artery wall could become thicker without reducing blood flow and so we have to ignore "thick".

Identify the missing words, A, B and C, in Figure 2.

(3)

A Carbon monoxide
B smooth muscle deposits cholesterol
C thick

This response was awarded 2 marks. Marks are awarded for correct responses for B and C. For A we can ignore reference to fats (this is too vague). For B plaque is the correct response - we can accept phonetic spellings and singular or plural. For C any word which clearly implies that the lumen (the inside of the artery we are referring to in the question as blood flow is reduced) can be accepted as "other relevant response" and so blocked can be awarded a mark.

(3)

A fat
fatty
B blockage Plaque
C blocked

This response was awarded 3 marks. An acceptable missing word is given for A, B and C. For B plaques is the correct response but can accept "fatty deposits".

A Cholesterol
B fatty deposits
C Blocked

Q2(a)

This was a 3 mark question.

The conversion mark was rarely awarded, leading many learners to get 2 marks for an incorrect answer. Most learners recalled the equation and applied it well to obtain partial credit even if they failed to provide the correct answer.

This response was awarded 1 Mark.

The learner has correctly calculated the conversion and so 1 mark can be awarded. However, they later reverse the conversion by multiplying by 1000 - and so this mark does not get given. Unfortunately, the learner has inverted the division - but their evaluation is correct and so 1 mark can be awarded overall. When the response isn't 400 or a 4 to the power of ten POT error response then the working has to be investigated to see if marks can be awarded.

(3)

$$\text{mag} = \frac{\text{actual size}}{\text{image size}}$$

$$1 \text{ mm} = 1000 \mu\text{m}$$

$$13.2 \times 1000 = 13,200$$

$$\frac{33}{13,200} = 0.0025 \times 1000 = 2.5$$

magnification = 2.5 mm x

This response was awarded 3 marks.

With no answer on the response line the 400 at the end of the learners working is accepted and so 3 marks can be awarded. The unit is given at the end of the response line and as there is not a mark allocated for the recall of the unit the unit that the learner has provided can be ignored.

(3)

M A

$$\text{Magnification} = \text{Image size} \div \text{Actual size}$$

$$13.2 \text{ mm} \times 1000 = 13200 \mu\text{m}$$

$$13200 \mu\text{m} \div 33 = 400 \mu\text{m}$$

~~400~~

magnification = x

Q2(b)(i)

This was a 2 mark question.

Most marks were awarded for mp1 and mp2, occasionally mp7, but unfortunately the other marking points were rarely seen. The majority of learners were able to obtain at least 1 out of the 2 marking points. Common correct responses included the ideas about observing live specimens, no specialist training required, and portability.

Common mistakes came from a lack of detail, for example, giving an answer of "easier" or "small" with no reference to portability, and so repeating the points given in the stem.

This response was awarded 1 mark.

The response on line 1 is creditworthy for MP1 and so 1 mark is awarded.

The "run on low electricity" is a common response and is a little too vague to be creditworthy.

(b) Two advantages of light microscopes compared to electron microscopes are:

- Light microscopes are cheaper.
- Light microscopes are easier to use.

(i) Give **two other** advantages of using a light microscope. (2)

1 Specimens can be alive or dead

2 Run on low electricity

This response was awarded 2 marks.

The learner has given 2 creditworthy responses and so scores maximum marks - the light microscope can be used to observe living cells and is portable.

(b) Two advantages of light microscopes compared to electron microscopes are:

- Light microscopes are cheaper.
- Light microscopes are easier to use.

(i) Give **two other** advantages of using a light microscope. (2)

1 the light microscope can observe living cells
whereas the electron microscope cannot

2 ~~you can bring a light mic~~ it is portable
whereas and easy to carry to a different place

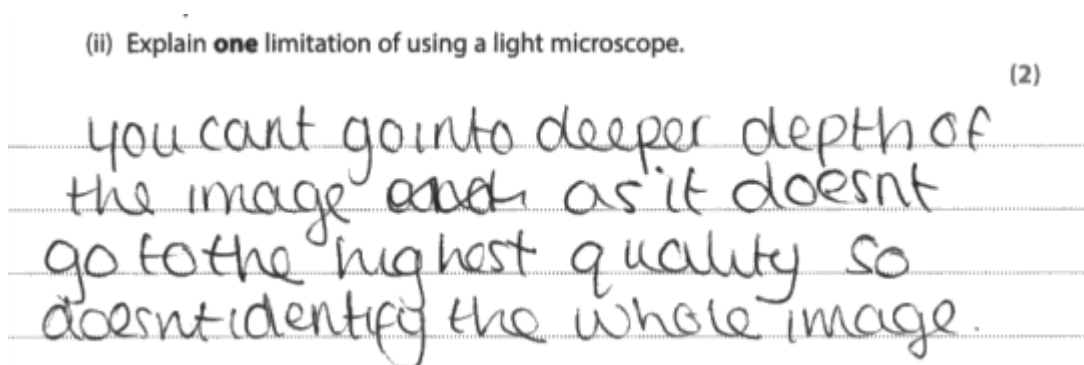
Q2(b)(ii)

This was a 2 mark question. Most learners obtained one or more of MP1, MP2 or MP3, but the other marking points were rarely seen. Many learners were able to identify that light microscopes have lower magnification (often referred to as 'zoom') and link this to resolution.

The scientific language used was weak. There were almost no mentions of "resolving power", wavelength or depth of field.

This response was awarded 1 mark.

This idea is rarely seen and sits outside the identification points surrounding magnification and resolution limitations on the mark scheme. The learner has given MP5 as their identification point and so can be credited with one mark. Unfortunately, the expansion given with this identification point should refer to the diffraction of light and so the learner's comment does not identify the whole image and is not sufficient for the second mark in the pair.



This response was awarded 2 marks.

The learner has used wavelength of light MP4 as the identification point of their explanation - this marking point is not seen often. They have given a linked expansion point - that less detail (MP3) is seen as a result of the greater wavelength of light. The learner's wording of "less detail to be seen" is just sufficient for MP3.

(f) Explain ~~one~~ limitation of using a light microscope.

(2)

Wavelength is a limiting factor of light microscopes. They have a greater wavelength than electron microscopes which as a result allows less detail to be seen.

Q3(a)

This was a 1 mark question. This was a basic recall question, and most learners were able to give a correct response.

This response was awarded 0 marks.

myofibrilament nucleus.

This response was awarded 1 mark.

myosin

Q3(b)

This was a 1 mark multiple choice question and was difficult. The learner was required to demonstrate a detailed understanding of organelles to select the correct pair.

This response was awarded 1 mark.

Identify the pair of organelles that both have a double membrane.

- ☐ A mitochondrion and lysosome
- ☐ B mitochondrion and ribosome
- ☒ C nucleus and mitochondrion
- ☐ D nucleus and ribosome

Q3(c)

This was a 2 mark question.

The second marking point was seen most commonly, mainly through some variation of “pumps blood around the body”. Some were able to gain the second mark for the idea that the heart is an organ as it is a “collection of tissues”, although it was clear that the learners only had a very general idea, and the scientific language was very weak. Common mistakes for this question arose from a lack of detail, for example, ‘we need it to function’.

This response was awarded 1 mark.

Basically, we are looking for 2 parts to the response - they do not have to be on the numbered lines - both marks could be achieved from 1 statement, but the numbers have been given as scaffolding. Here MP1 has been awarded as the learner has given a named tissue (muscle) and as this is plural, we can give benefit of the doubt that this is sufficient for made up of different tissues. Unfortunately, MP2 - about having a specific function - is too vague for a level 3 definition. Contributing to everyday life, allowing survival, being part of the body, etc are too vague for the function and so the second mark cannot be awarded.

(c) The heart is an example of an organ.

Give **two** reasons why the heart is called an organ. (2)

1. Because it has muscles in it just like a lung ~~recess~~

2. Another reason is that the heart contributes to your everyday life, so it is therefore classed as a muscle.

(Total for Question 3 = 4 marks)

This response was awarded 2 marks.

The learner has given both marking points (and in fact has given 3 correct responses) and so maximum marks can be awarded for this question.

- 1 Built of specialised tissues
- 2 Carries out a specific function, being part of an organ system

Q4(a)

This was a 1 mark question. This was a recall question with at least half of the learners giving the correct response. Where an incorrect response was given it was usually potassium or calcium.

This response was awarded 0 marks.

(1)

potassium

This response was awarded 1 mark.

(1)

Sodium

Q4(b)

This was a 2 mark question. This question was poorly answered overall. The most common marking point was for an idea of re-synthesis or 'repeat' in relation to serotonin, although the scientific language used was once again poor. There was some reference to the regulation of action potentials by stronger learners.

Key terminology such as receptors, depolarisation or action potential were rarely seen, and many answers showed a lot of confusion and lack of understanding.

This response was awarded 1 mark. This response is describing, in some detail, further action potentials and so MP4 - enables the process to repeat - can be awarded. This is a quite common idea within the responses but is worded in a variety of ways.

(4)

they are reabsorbed so that when the next action potential arrives at the presynaptic ^{neurone} ~~membrane~~, the neurotransmitters are ready to fuse with pre synaptic membrane and transfer the action potential to the post synaptic neurone.

This response was awarded 2 marks.

The learner's first comment about vesicles can be ignored. Their reference to receptors is sufficient for MP1. They then describe that if serotonin is left depolarisation will continue (through a description) and so that is sufficient for MP2.

So the vesicles can transport them
to the presynaptic membrane in order
to fuse with it so the neurotransmitter
can be released. If the serotonin is left
on the receptor the sodium ion gated
channels will remain open causing sodium ions
to continue entering. Reabsorbing the neurotransmitters
will close the gates.

Q4(c)

This was a 4 mark question.

This question was also answered poorly with a lot of confusion and poor comprehension of the topic. Most of the creditworthy answers received marks for MPs 6 & 7. Very few learners managed to obtain any of the other available marking points and scientific language was again poor.

There was a common misconception that drug F would increase serotonin levels and thus, make the person happier.

This response was awarded 1 mark.

The learner has stated the serotonin is going to be present for longer within the synapse this is too vague. For MP1 the detail of staying in the cleft, in the gap of the synapse or between the neurones would be sufficient, but synapse alone is too vague. However, the serotonin is linked to feeling more content - which is sufficient for MP7.

(4)

Drug F not allowing the serotonin to be reabsorbed means that the serotonin is going to be present for longer within the synapse. If serotonin is present for longer then the patient will feel a lot more content, which is a result of serotonin being present

This response was awarded 4 marks.

The learner has given MP1 with their statement that there will be increased serotonin level in the synaptic cleft - they have linked this to the outcome of the drug - a feeling of contentment and pleasure (MP7). The learner then adds more detail that serotonin will bind to the receptors (MP2) in the postsynaptic membrane (MP3). "On the postsynaptic membrane" is an additional mark as this provides extra detail. The "continue on to the next neurone" is not sufficient alone for MP4, but this learner has already been awarded maximum marks for the question.

Preventing the reabsorption of serotonin at the synapse helps increase the level of ~~the~~ serotonin, as it is not reabsorbed. Increasing the serotonin levels in the synaptic cleft and between neurones will improve the patient's ~~mood~~ mood by giving them contentment and pleasure. The ~~3~~ many serotonin will bind to the serotonin receptors in the postsynaptic ~~neuron~~ membrane to ~~send~~ and continue on to the next neurone.

Q5

This was a 6 mark extended open response question.

On the whole, learners were clearly more comfortable discussing neutrophils rather than lymphocytes, the structure and function of which were rarely mentioned. Some learners showed a poor understanding of the function and structure of each cell. Those learners who were able to provide creditworthy answers obtained marks for noting the similarities in structure and function of the cells.

A common misconception seen several times described one cell as eukaryotic and the other as prokaryotic.

This response was awarded 2 marks.

Responses in tables or spider diagrams or any layout is acceptable.

Unfortunately, there are a number of incorrect statements in the learner's comparison table neutrophil being large and the lymphocyte is a correct difference. The organelles listed as "they both have" after the table are correct. On balance, with the other statements being incorrect and so 2 marks have been awarded.

(6)

Neutrophil	Lymphocyte
not a blood cell	not white blood cell
has many lots of membrane bound organelles	- doesn't have many membrane bound organelles
- large size	- small size
- has has a vacuole, chloroplast	- digests/engulfs pathogens
	- produces antibodies to fight pathogens
	- mitochondria monoclonal
	- circular shaped antibodies
- Both have cytoplasm	
- Both have plasma membrane	
- Both have nucleus	

This response was awarded 4 marks.

The learner has given several correct statements - the neutrophil has a lobed nucleus, the neutrophil "wraps around a pathogen", a lymphocyte sends out chemical signals, the lymphocyte is smaller. However, the lines of argument are incomplete with some detail omitted - such as reference to antibodies, types of lymphocytes, lysosomes in neutrophils... Similarities of both have a nucleus and a cell membrane have been given. They have hinted that both are involved in defence against pathogens but have not said they are both white blood cells. On balance this is a borderline level 2/ level 3 response with 4 marks awarded.

A neutrophil has a lobed nucleus whereas a lymphocyte has not. This is because a neutrophil is meant to wrap around a pathogen to neutralise it ~~but~~ but a lymphocyte is meant to send out chemical signals to neutrophils so that they know where to attack.

The lymphocyte is smaller because it doesn't need to be able to wrap around pathogens like a neutrophil. Both lymphocytes and neutrophils have a nucleus that stores genetic information and a cell membrane to stop things from getting into the cell.

The response below was awarded 5 marks.

The learner has given a number of correct statements. Ideally there would be similarities and differences for a level 3 response, but they don't have to be equally balanced (e.g. 5 differences and 1 similarity could still be a level 3 response). There are attempts to link structure to function - again a level 3 style response. The learner has covered the majority of the indicative content - they do not have to cover it all and they may have other relevant responses which are creditworthy. However, there are some incorrect statements which means 6 marks would not be awarded.

- neutrophil and lymphocytes are both types of white blood cells
- two types of lymphocytes - B lymphocytes and T lymphocytes
- there is only one neutrophil
- ~~• neutrophil and white blood cells are used to target pathogens~~
- neutrophil and lymphocytes are used to target pathogens that enter the body
- neutrophil has a lobed nucleus because it is needed to make more of that neutrophil. Whereas lymphocytes have antigens around them so they are able to attract the pathogen as they are complementary and fight the pathogen.
- neutrophils are used to fight bacteria that have not yet entered into the blood stream and so the lobed nucleus makes more of them so that it ~~can~~ can kill the ^{pathogens} ~~bacteria~~. lymphocytes are used to target ~~bacteria~~ pathogens that have entered the blood stream.
- B lymphocytes are used first and the T lymphocytes are used after.
- lymphocytes memorise the pathogen that was in the body, so when it appears again it is ready to attack. ~~Neutrophils only attack once~~
- lymphocytes have lysosome which is used to break down the pathogen and remove the digested pathogens.

Summary

Based on their performance on this paper, learners should:

- Understand the demand of the command verbs, especially the difference between describe and explain
- Be familiar with and recognise the ultrastructure and function of organelles in the following cells: prokaryote cells (bacterial cells), eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific)
- Revise key definitions with level 3 detail, such as organ, synapse, neurotransmitter, etc.
- Show their working out for all calculations to ensure marks can be given for the stages within the calculation even if the final answer is incorrect
- Use the additional guidance to understand the depth and breadth of the specification